heavy cavalry, and infantry broke them down. Napoleon's approach on land was mirrored at sea by the British navy, dedicated to swift and decisive engagement with the enemy whenever feasible. The fact that Napoleon was ultimately defeated did nothing to tarnish the reputation of the methods with which he had won so many outstanding victories. European armies were dedicated to fighting large-scale battles to the death – ensuring massive casualties, as at Borodino in 1812, when more than 70,000 men were killed or wounded in a day.

TECHNOLOGICAL DEVELOPMENTS

The potential destructiveness of warfare was increased through the 19th century by rapid developments in many fields of technology. The accuracy, range, and rate of fire of infantry weapons were transformed as muzzle-loading muskets were superseded by breech-loading repeater rifles and then, from the 1880s, by fully automatic machine-guns. Rifled artillery replaced smoothbore cannon and fired high-explosive shells at long range. Industrialized countries were able to produce such arms and munitions in unprecedented quantities. Meanwhile, the construction of railways allowed large armies to move at unprecedented speed – although once away from the tracks they reverted to the pace of the march and the horse. The invention of the electric telegraph gave some help in controlling large-scale forces operating over wide geographical areas.

The American Civil War (1861–65) occurred before technological changes in weaponry were very far advanced. But even with muzzle-loading rifles and smoothbore cannon, the Union and Confederate armies amply demonstrated the potential for slaughter on battlefields where, counter to the aspirations of Napoleonic warfare, massed firepower was always likely to give the defence the upper hand. The cavalry charge in particular was shown to be outdated as it was simply unsustainable in the face of concentrated infantry and artillery fire.

While modern weapons technology was increasing the soldier's chances of dying on the battlefield, other manifestations of progress made his life substantially easier in many respects. The development of canned food, first issued to troops during the Napoleonic Wars, was a boon for men on campaign – opening cans became, for a time, the main use for bayonets. Mass production in general ensured that soldiers from industrialized countries were better clothed and shod than their rural ancestors.

In line with social reforms throughout the industrialized world, military punishments tended to become less harsh – flogging was officially abolished in the British and US armies in the 1860s. The pioneering activities of British nurse Florence Nightingale during the Crimean War (1854–56) and the founding of the International Red Cross in 1863 exemplified a new concern for the treatment of the war-wounded. Improvements in sanitation and in medical knowledge ensured that by the 20th century, for the first time in history, armies at war might potentially experience more deaths from actual combat than from the hardship and disease that accompanied any prolonged campaign.



US Civil War
A young drummer boy of the Union army poses in front of a cannon. Some drummer boys were as young as seven or eight. The most widely used artillery piece of the Civil War was the bronze 12-pounder field gun known as a Napoleon cannon. Most cannons of the Civil War were smoothbores with a range of only about 1.6km (1 mile).

IMPERIAL POWER

With their industrial might, increasing populations, and aggressive attitude to military operations, 19th-century Europeans and their overseas offshoots enjoyed a clear military superiority over non-European states. The Chinese government was forced to adhere to humiliating treaties through a series of military defeats, and the British took possession of India with piecemeal victories over local armies. European military expansion was certainly not wholly uncontested, and encounters between European armies and peoples with an indigenous warrior tradition were at times fought on surprisingly equal terms. The Maori in New Zealand, having transformed their traditional warfare by the adoption of firearms, at times outfought white settlers and a colonial army. The Zulu, one of many African peoples who resisted European imperial takeover in the late 19th century, were able on several occasions to bring the British infantry to close-quarters combat at which their spearmen excelled. In North America, the Plains Indians fought a guerrilla war against the US cavalry which was only suppressed with some difficulty. However, if an industrialized power had the will to devote all the necessary resources to a conflict, there could be no doubt as to the eventual outcome.

By the end of the 19th century, a wide technological divide had opened up between the world's major powers and the rest. The traditional warrior virtues of courage and aggression were of little use against the machine-gun. At the battle of Omdurman in 1898, the British army lost only 48 men while slaughtering some 16,000 followers of the Sudanese Mahdi. The English poet Hilaire Belloc was inspired to write the cruel couplet: "Whatever happens we have got/ The Maxim gun and they have not." But in 1914–18 the industrial powers would instead turn their destructive firepower upon one another, bringing the slaughter home to Europe.





